
PH102: Physics-2

Electrodynamics

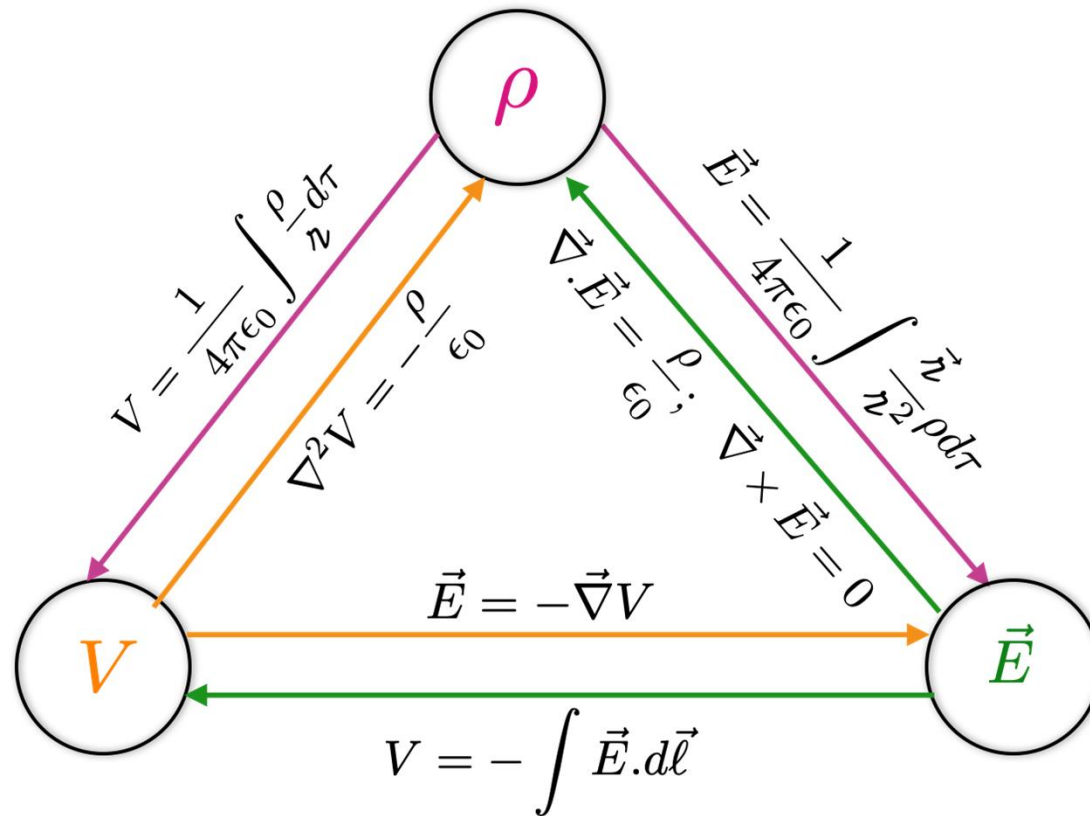
Lecture # 07

Electrostatics – Work and energy, and
Capacitance
31.01.2024

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Previous lecture

E , V and ρ are important parameters in electrostatics.



The work done to move a charge

In the previous lecture we calculated the work done by you to move a test charge Q from point $\mathbf{a}$ to point $\mathbf{b}$.

What we're interested in is the *minimum force you must exert* to do the job?

(Why?)

The work is therefore

$$W = \int_a^b \mathbf{F} \cdot d\mathbf{l} = -Q \int_a^b \mathbf{E} \cdot d\mathbf{l} = Q[V(\mathbf{b}) - V(\mathbf{a})]$$

Dividing by $Q \Rightarrow V(\mathbf{b}) - V(\mathbf{a}) = \frac{W}{Q}$

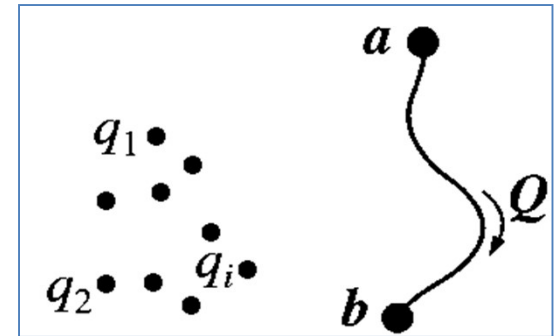
In particular, if you want to bring the charge Q in from far away and stick it at point $\mathbf{r}$, the work you must do is

$$W = Q[V(\mathbf{r}) - V(\infty)]$$

i.e., If you have set the reference point at infinity,

$$W = QV(\mathbf{r})$$

i.e., Potential is the potential energy (the work it takes to create the system) per unit charge (just as the field is the force per unit charge)



The energy of a point charge distribution

How much work would it take to assemble an entire *collection of point charges*?

Imagine bringing in the charges, one by one, from far away as shown in the figure.

The first charge, q_1 , takes no work, since there is no field yet to fight against.

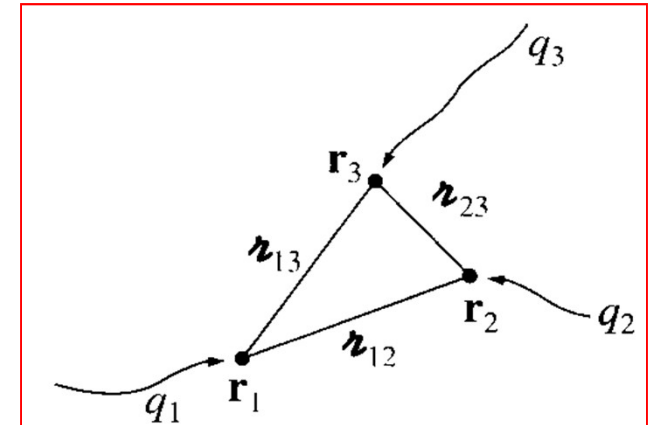
Now to bring q_2 , followed by q_3 , we have

$$W_1 = 0, \quad W_2 = \frac{1}{4\pi\epsilon_0} q_2 \left(\frac{q_1}{\lambda_{12}} \right), \quad W_3 = \frac{1}{4\pi\epsilon_0} q_3 \left(\frac{q_1}{\lambda_{13}} + \frac{q_2}{\lambda_{23}} \right)$$

$$W = \frac{1}{4\pi\epsilon_0} \left(\frac{q_1 q_2}{\lambda_{12}} + \frac{q_1 q_3}{\lambda_{13}} + \frac{q_2 q_3}{\lambda_{23}} \right)$$

$$W = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \sum_{\substack{j=1 \\ j>i}}^n \frac{q_i q_j}{\lambda_{ij}} = \frac{1}{8\pi\epsilon_0} \sum_{i=1}^n \sum_{\substack{j=1 \\ j \neq i}}^n \frac{q_i q_j}{\lambda_{ij}}$$

$$= \frac{1}{2} \sum_{i=1}^n q_i \left(\frac{1}{4\pi\epsilon_0} \sum_{\substack{j=1 \\ j \neq i}}^n \frac{q_j}{\lambda_{ij}} \right) = \frac{1}{2} \sum_{i=1}^n q_i V(\mathbf{r}_i)$$



$j > i$ ensures that the pairs are not counted twice, otherwise introduce a factor of half. Avoid $i = j$.

The term within the parenthesis is the potential at point $\mathbf{r}_i$ (the position of q_i) due to all other charges.

This is the work done to assemble or dismantle a charge configuration.

The energy of a continuous charge distribution

The work it takes to assemble a configuration of point charges is $W = \frac{1}{2} \sum_{i=1}^n q_i V(\mathbf{r}_i)$

For a volume charge density ρ , the work done is $W = \frac{1}{2} \int \rho V d\tau$

For a line charge density λ , the work done is $W = \frac{1}{2} \int \lambda V d\mathbf{l}$

For a surface charge density σ , the work done is $W = \frac{1}{2} \int \sigma V d\mathbf{a}$

W can be expressed in term of **E** as follows

$$dW_i = \frac{1}{2} (dq_i) V_i(\mathbf{r}_i) = \frac{1}{2} \rho_i V_i(\mathbf{r}_i) d\tau$$

$$W = \frac{1}{2} \int \rho V d\tau = \frac{1}{2} \int (\epsilon_0 \nabla \cdot \mathbf{E}) V d\tau = \frac{\epsilon_0}{2} \int (\nabla \cdot \mathbf{E}) V d\tau$$

$$\nabla \cdot (V \mathbf{E}) = (\nabla V) \cdot \mathbf{E} + (\nabla \cdot \mathbf{E}) V \quad \text{Integral by parts}$$

$$W = \frac{\epsilon_0}{2} \int (\nabla \cdot \mathbf{E}) V d\tau = \frac{\epsilon_0}{2} \left[- \int \mathbf{E} \cdot (\nabla V) d\tau + \oint (V \mathbf{E}) \cdot d\mathbf{a} \right]$$

$$W = \frac{\epsilon_0}{2} \left[\int E^2 d\tau + \oint (V \mathbf{E}) \cdot d\mathbf{a} \right]$$

$$W = \frac{\epsilon_0}{2} \int_{\text{all space}} E^2 d\tau$$

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Gauss's law

Divergence theorem

The integration is only within the region where charges are present. Extending the integration over entire space, the surface integral goes to zero as r goes to infinity.

Comments on electrostatic energy

- A perplexing “inconsistency”?

$$W = \frac{1}{2} \sum_{i=1}^n q_i V(\mathbf{r}_i) > 0 \text{ or } < 0$$

(Energy for the fabrication of the point charges is not taken into account. *We started with point charges and simply found the work required to bring them together.*)

whereas $W = \frac{\epsilon_0}{2} \int E^2 d\tau > 0$!!! (Total energy stored in the configuration.)

The inconsistency is between the expressions

$$W = \frac{1}{2} \sum_{i=1}^n q_i V(\mathbf{r}_i) \quad \text{and} \quad W = \frac{1}{2} \int \rho V d\tau$$

$V(\mathbf{r}_i)$ represents the potential due to all the *other* charges *but not* q_i , whereas in the latter, $V(\mathbf{r})$ is the *full* potential. For a *continuous* distribution, there is no distinction, since the amount of charge *right at the point* $\mathbf{r}$ is vanishingly small, and its contribution to the potential is zero.

➤ Where is the energy stored?

$W = \frac{1}{2} \int \rho V d\tau$ suggests that the energy is stored in the charge distribution.

$W = \frac{\epsilon_0}{2} \int E^2 d\tau$ suggests that the energy is stored in the field $\mathbf{E}$.

Though one expression is derived from the other but they involve different regions (e.g., spherical shell: charge is on the surface but the field extends out beyond the surface).

In electrostatics, we say that the energy is stored in the charge. In the context of radiation theory, it is stored in the field.

➤ Electrostatic energy does not obey superposition principle because it is quadratic in fields.

$$\begin{aligned} W_{\text{tot}} &= \frac{\epsilon_0}{2} \int E^2 d\tau = \frac{\epsilon_0}{2} \int (\mathbf{E}_1 + \mathbf{E}_2)^2 d\tau = \frac{\epsilon_0}{2} \int (E_1^2 + E_2^2 + 2\mathbf{E}_1 \cdot \mathbf{E}_2) d\tau \\ &= W_1 + W_2 + \epsilon_0 \int \mathbf{E}_1 \cdot \mathbf{E}_2 d\tau. \end{aligned}$$

e.g., doubling the charge will quadruple the energy.

Conductors

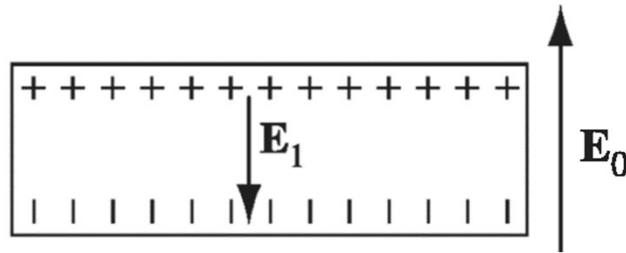
In a metallic **conductor**, one or more electrons per atom are free to roam.

A perfect conductor would contain an unlimited supply of free charges.

In real life there are no perfect conductors (metals come pretty close).

Properties:

- **$\mathbf{E} = 0$ inside a conductor:** Field due to induced charges in a conductor cancels the originally applied field.



- **$\rho = 0$ inside a conductor:** $\nabla \cdot \mathbf{E} = \rho/\epsilon_0 = 0$ if $\mathbf{E} = 0$. There is still charge around, but exactly as much plus charge as minus, so the net charge density in the interior is zero.

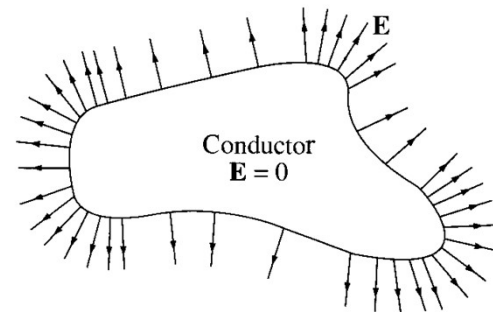
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- **Any net charge resides on the surface:** That's the only other place it can be. Essentially, the charge on a conductor will seek the configuration that minimizes its potential energy (with specified shape and total charge).

For a uniformly charged spherical shell, $W = \frac{1}{8\pi\epsilon_0} \frac{q^2}{R}$

For a uniformly charged sphere, $W = \frac{3}{20\pi\epsilon_0} \frac{q^2}{R}$

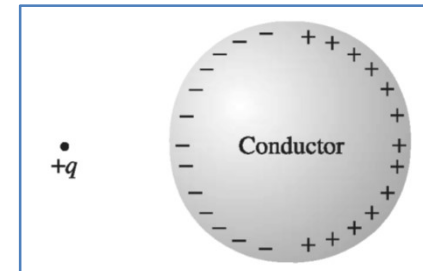
- **A conductor is an equipotential:** $V(\mathbf{b}) - V(\mathbf{a}) = - \int_a^b \mathbf{E} \cdot d\mathbf{l} = 0$ within (or at the surface of) a given conductor and hence $V(\mathbf{a}) = V(\mathbf{b})$.
- **$\mathbf{E}$ is perpendicular to the surface, just outside a conductor:** If there is any tangential component, the charges will flow around the surface until, it kills off the tangential component of $\mathbf{E}$. The charges cannot flow perpendicular to surface, as they are confined within the conductor.



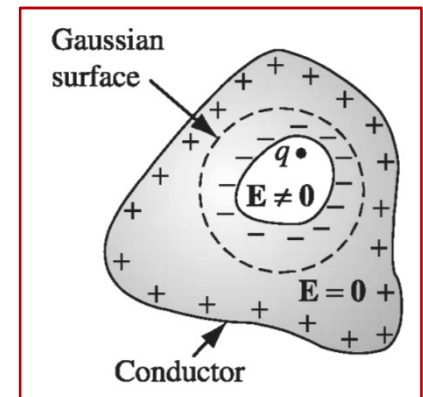
Induced charges

An external field or a charge may influence the charges in a material medium and thus allow them to move and redistribute within or on the surface of the medium.

- Negative charges move closer to the $+q$ charge and positive charges move away.
- There is a net force of attraction.



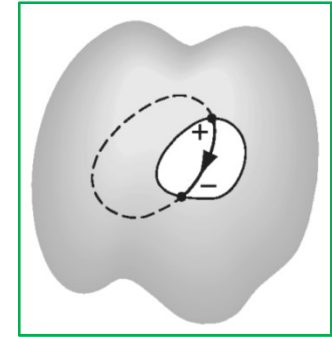
- For a charge inside a cavity in a conductor, $\mathbf{E} \neq 0$.
- $\mathbf{E} = 0$ within the conductor. So, there are induced negative charges closer to the cavity but just inside the conductor at its boundary.
- The compensating charge on the *outer* surface "communicates" the presence of q to the outside world.
- The cavity and its contents are electrically isolated from the outside world by the surrounding conductor.



For all points in the conductor $\oint \mathbf{E} \cdot d\mathbf{a} = 0$;
 $Q_{\text{enc}} = q + q_{\text{induced}} = 0$,
so $q_{\text{induced}} = -q$.

(Contd...)

- $\mathbf{E}$ is zero within the conductor.
- For an empty cavity inside a conductor, $\mathbf{E} = 0$ (if at all there is a field within the cavity then $\oint \mathbf{E} \cdot d\mathbf{l} > 0$ violating the condition for an electrostatic field that $\oint \mathbf{E} \cdot d\mathbf{l} = 0$).



This is why you are relatively safe inside a metal car during a thunderstorm - you may get *cooked*, if lightning strikes, but you will not be *electrocuted*.

The same principle applies to the placement of sensitive apparatus inside a grounded **Faraday cage**, to shield out stray electric fields.

- ✓ Microwave oven.
- ✓ Magnetic Resonance Imaging (MRI) scanners.
- ✓ Mobile phone wrapped in a plastic and then by an aluminum foil.



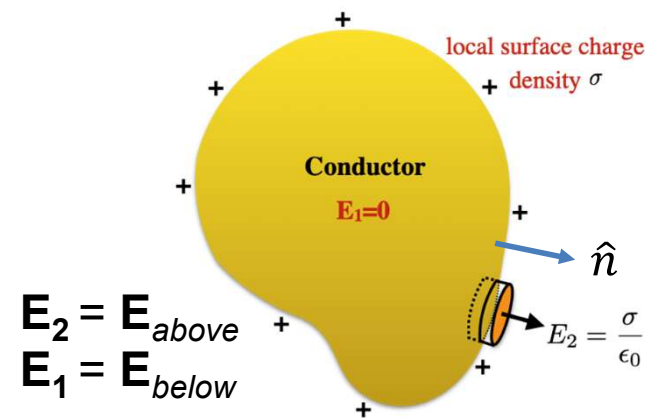
<https://www.sciencefacts.net/faraday-cage.html>

Surface charge and the force on a conductor

Across a boundary, in general, $\mathbf{E}_2 - \mathbf{E}_1 = \frac{\sigma}{\epsilon_0} \hat{n}$

For a conductor, $\mathbf{E}_1 = 0$, therefore, $\mathbf{E}_2 = \frac{\sigma}{\epsilon_0} \hat{n}$

Also, $\left(\frac{\partial V_2}{\partial n} - \frac{\partial V_1}{\partial n} \right) = -\frac{\sigma}{\epsilon_0} \Rightarrow \sigma = -\epsilon_0 \frac{\partial V_2}{\partial n}$



Thus, the surface charge on a conductor can be calculated if $\mathbf{E}$ or V is known.

Force experienced by a charge is $\mathbf{F} = q\mathbf{E}$.

Force experienced by a surface charge is $\mathbf{F} = \sigma A \mathbf{E}$

Therefore, the force per unit area, $\mathbf{f} = \sigma \mathbf{E}$.

$\mathbf{E}$ is discontinuous, which value are we supposed to use: $\mathbf{E}_2$, $\mathbf{E}_1$, or something in between?

Use the *average* of the two:

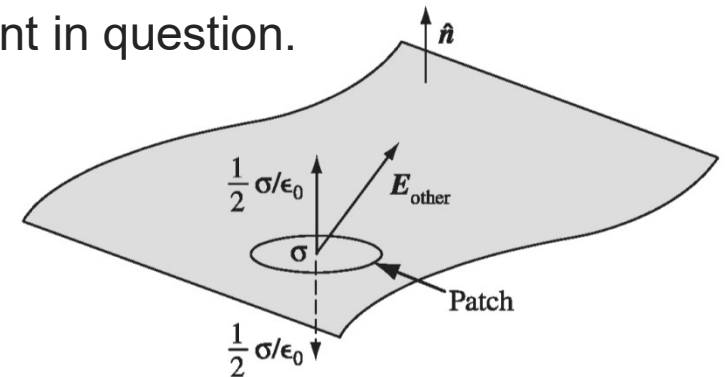
$$\mathbf{f} = \sigma \left(\frac{\mathbf{E}_{above} + \mathbf{E}_{below}}{2} \right)$$

Why average?

(Contd...)

Consider a tiny patch of surface surrounding the point in question. The *total* field has two parts – due to the patch itself and that due to everything else (other region of the surface and any other source),

$$\mathbf{E} = \mathbf{E}_{\text{patch}} + \mathbf{E}_{\text{other}}$$



The force on the patch is due to $\mathbf{E}_{\text{other}}$, it cannot exert force onto itself.

$$\mathbf{E}_{\text{patch}} = \pm \frac{\sigma}{2\epsilon_0} \hat{n}$$

We can write, $\mathbf{E}_{\text{above}} = \mathbf{E}_{\text{other}} + \frac{\sigma}{2\epsilon_0} \hat{n}$; $\mathbf{E}_{\text{below}} = \mathbf{E}_{\text{other}} - \frac{\sigma}{2\epsilon_0} \hat{n}$.

Therefore, $\mathbf{E}_{\text{other}} = \left(\frac{\mathbf{E}_{\text{above}} + \mathbf{E}_{\text{below}}}{2} \right) = \mathbf{E}_{\text{average}}$

So, the force per unit area is $\mathbf{f} = \frac{1}{2\epsilon_0} \sigma^2 \hat{n}$

This gives rise to an outward **electrostatic pressure** on the surface, $P = \frac{1}{2} \epsilon_0 E^2$

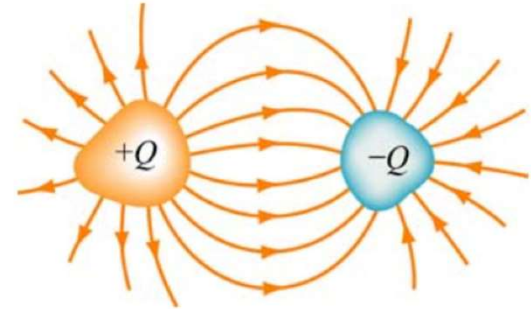
Capacitor

The potential difference between two conductors and having charges $+Q$ on one and $-Q$ on the other is

$$V = V_+ - V_- = -\int_+^- \mathbf{E} \cdot d\mathbf{l}$$

We know that $\mathbf{E}$ is proportional to Q given by

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho}{r^2} \hat{r} d\tau$$



Since $\mathbf{E}$ is proportional to Q , so also is V . The constant of proportionality is called the capacitance of the arrangement,

$$C = Q/V$$

Capacitance is a purely geometrical quantity, determined by the sizes, shapes, and separation of the two conductors.

In SI units, C is measured in farads (F); a farad is a coulomb-per-volt.

(Contd...)

How much work does it take, then, to charge the capacitor up to a final amount Q ?

Suppose that at some intermediate stage in the process the charge on the positive plate is q , so that the potential difference is q/C .

The work you must do to transport the next piece of charge, dq , is

$$dW = \left(\frac{q}{C} \right) dq$$

The total work necessary, then to go from $q = 0$ to $q = Q$, is

$$W = \int_0^Q \left(\frac{q}{C} \right) dq = \frac{1}{2} \frac{Q^2}{C}$$

Since $Q = CV$

$$W = \frac{1}{2} CV^2$$

where V is the final potential of the capacitor.

Solved example 1

Find the energy of a uniformly charged spherical shell of total charge q and radius R .

Method 1

We know that $W = \frac{1}{2} \int \sigma V da$.

The potential at the surface of this sphere is $\frac{1}{4\pi\epsilon_0} \frac{q}{R}$

Therefore, $W = \frac{1}{8\pi\epsilon_0} \frac{q}{R} \int \sigma da = \frac{1}{8\pi\epsilon_0} \frac{q^2}{R}$.

Method 2

Inside the sphere, $\mathbf{E} = 0$; and outside,

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{\mathbf{r}}, \quad \text{so} \quad E^2 = \frac{q^2}{(4\pi\epsilon_0)^2 r^4}.$$

$$W_{\text{tot}} = \frac{\epsilon_0}{2(4\pi\epsilon_0)^2} \int_{\text{outside}} \left(\frac{q^2}{r^4} \right) (r^2 \sin\theta dr d\theta d\phi) = \frac{1}{32\pi^2\epsilon_0} q^2 4\pi \int_R^\infty \frac{1}{r^2} dr = \frac{1}{8\pi\epsilon_0} \frac{q^2}{R}.$$

Solved example 2

Find the capacitance of two concentric spherical metal shells, with radii a and b .

Place charge $+Q$ on the inner sphere and $-Q$ on the outer one. The field between the sphere is

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2} \hat{\mathbf{r}}$$

The potential difference between the plates is

$$V = - \int_b^a \mathbf{E} \cdot d\mathbf{l} = - \frac{Q}{4\pi\epsilon_0} \int_b^a \frac{1}{r^2} dr = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{a} - \frac{1}{b} \right).$$

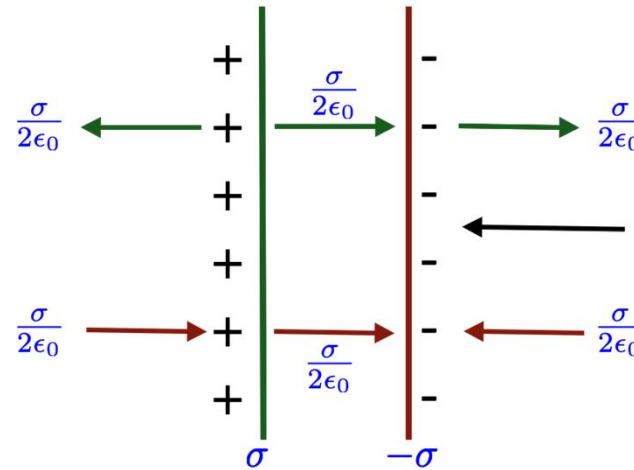
Since V is proportional to Q and so the capacitance is

$$C = \frac{Q}{V} = 4\pi\epsilon_0 \frac{ab}{(b-a)}. \quad \text{(Geometrical quantity)}$$

Solved example 3

Capacitance of a parallel plate capacitor with plates having charge densities $+\sigma$ and $-\sigma$, respectively.

Assumptions:
 Flat, parallel surfaces;
 distance d between plates is
 much smaller than $\sqrt{A}$ where
 A is the area of the plates



$$\sigma = Q/A$$

$$\vec{E} = \frac{\sigma}{\epsilon_0} \hat{z}$$

Constant field exists between the plates and elsewhere it is zero.

Therefore, $E = -\frac{dV}{dz} \Rightarrow V(z) = -Ez + \text{constant}$

Using the boundary conditions, $V(z = 0) = V$ and $V(z = d) = 0$, we get

$$V = Ed = \frac{\sigma d}{\epsilon_0} = \frac{Qd}{A\epsilon_0}$$

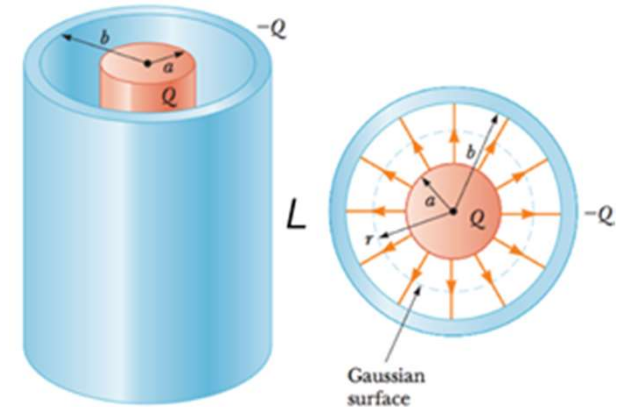
Therefore,

$$C = \frac{Q}{V} = \frac{\epsilon_0 A}{d}$$

(Geometrical quantity)

Solved example 4

A cylindrical capacitor has two co-axial cylinders of length L and radii a and b . The inner cylinder is given a charge of $+Q$ and the outer cylinder $-Q$. Determine the capacitance of the system. Neglect end effects (i.e., bending of field lines at the ends).



Choose the Gaussian surface as the coaxial cylinder.

$E = 0$ for $r > b$ and $r < a$ because $Q_{enc} = 0$.

For $a < r < b$

$$\oint_S \vec{E} \cdot d\vec{a} = EA = E2\pi rl = \frac{\lambda l}{\epsilon_0} \implies E = \frac{\lambda}{2\pi\epsilon_0 r}$$

$$\text{So, } \Delta V = V_b - V_a = - \int_a^b E_s ds = - \frac{\lambda}{2\pi\epsilon_0} \int_a^b \frac{dr}{r} = - \frac{\lambda}{2\pi\epsilon_0} \ln \left(\frac{b}{a} \right)$$

$$\text{Therefore, } C = \frac{Q}{|\Delta V|} = \frac{\lambda L}{\lambda \ln(b/a)/2\pi\epsilon_0} = \frac{2\pi\epsilon_0 L}{\ln(b/a)} \quad (\text{Geometrical quantity})$$

Questions?